

CoRoT-1b

R-2656

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_130114 · created 2026-08-29T23:51:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456306.7117 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.04
Transit depth (Rp/Rs)^2	2.01 +/- 1.12 [%]
Orbital Inclination (inc)	89.97 +/- 2.88
Airmass coefficient 1 (a1)	1.242 +/- 0.037
Airmass coefficient 2 (a2)	-0.109 +/- 0.018
Scatter in the residuals of the lightcurve fit is	2.95 %
Best Comparison Star	None
Optimal Aperture	3.57
Optimal Annulus	7.36
Transit Duration (day)	0.0632 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.04 vs 0.1388 (deviation 0.06σ)
Duration fit vs published	0.0632 d vs 0.1046 d (deviation 8.93σ)
Mid-time O-C	-5.6 min
Depth SNR	2.6
Residual scatter	2.95 %

Light curve

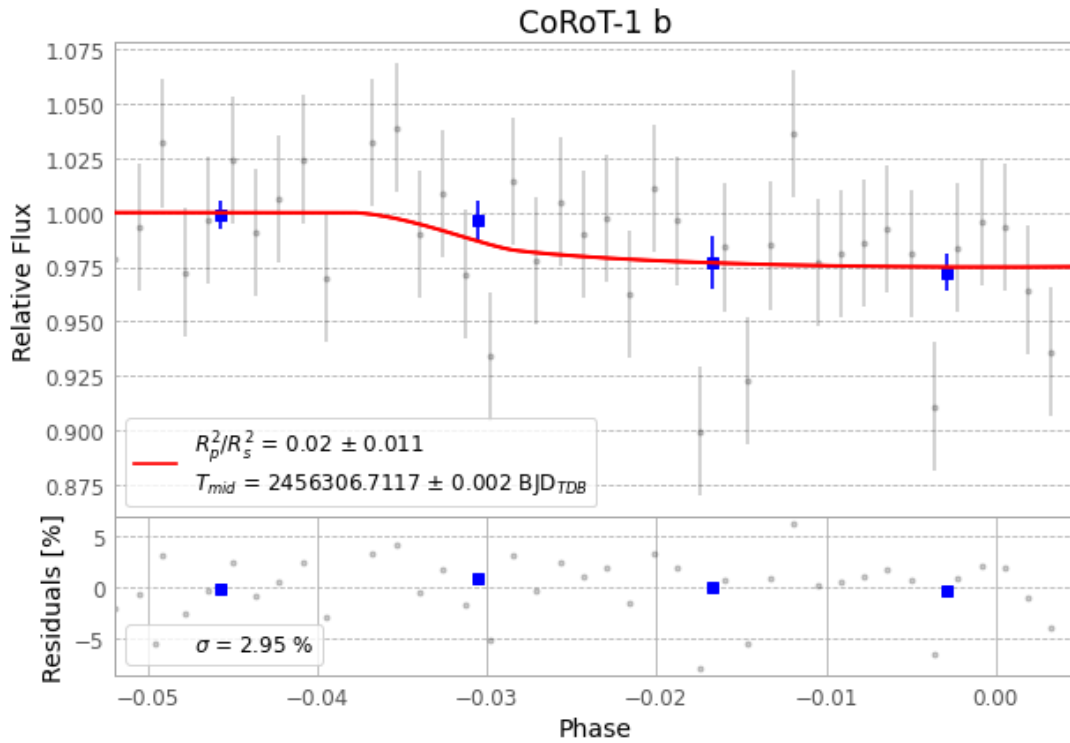


Figure 1 — FinalLightCurve_CoRoT-1 b_2013-01-13.png (SHA-256 29d07bf94844902e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [211, 235], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20db100b61e34588...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

42 FITS frames (28 MB), CoRoT-1130114030312.FITS → CoRoT-1130114050612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-130114.log (ee4fc41dd053...), FinalLightCurve_CoRoT-1 b_2013-01-13.png (29d07bf94844...), FinalLightCurve_CoRoT-1 b_2013-01-13.csv (adbf486e3572...)

Compute resources

Wall time 108.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:51Z.